

SANDSPIT, BC, Canada

WMO: 711110

Lat: 53.2494N

Lon: 131.8133W

Elev: 6

StdP: 101.25

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.3	-1.7	-10.5	1.5	-2.1	-8.1	1.9	-0.7	21.7	6.7	19.2	6.5	6.0	330	0.696

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.7	20.4	16.1	18.9	15.4	17.8	14.7	16.6	19.4	15.8	18.1	15.2	17.2	3.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	10.8	17.7	14.7	10.4	17.0	14.1	10.1	16.4	46.4	19.6	44.2	18.3	42.4	17.3	20.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.2	15.2	12.7	DB	-5.2	23.3	2.3	1.8	-6.9	24.6	-8.2	25.6	-9.4	26.6	-11.1	27.9	(4)
(5)			WB	-6.6	18.0	2.3	0.9	-8.2	18.7	-9.5	19.2	-10.8	19.7	-12.4	20.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	4.4	4.1	4.6	6.7	9.6	12.2	14.6	15.3	13.2	9.7	6.2	4.4	(6)	
	DBStd	4.62	2.88	2.37	2.41	1.75	1.96	1.85	1.77	1.63	1.75	2.04	2.55	2.70	(7)	
	HDD10.0	959	173	164	167	99	31	2	0	0	1	31	116	175	(8)	
	HDD18.3	3489	431	398	426	349	271	185	116	94	153	269	365	433	(9)	
	CDD10.0	515	0	0	0	1	19	67	144	166	98	20	1	0	(10)	
	CDD18.3	3	0	0	0	0	0	0	2	1	0	0	0	0	(11)	
	CDH23.3	4	0	0	0	0	0	1	2	1	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	5.0	5.7	5.5	5.3	5.1	4.9	4.6	4.1	4.1	4.4	5.1	5.4	5.7	(14)	
(15) Precipitation	PrecAvg	1273	158	119	104	89	57	53	41	53	85	175	178	166	(15)	
	PrecMax	2017	321	270	226	210	134	139	144	147	195	311	301	350	(16)	
	PrecMin	350	36	0	0	25	12	15	13	0	18	40	43	1	(17)	
	PrecStd	263	67	60	48	46	27	27	26	30	42	66	63	60	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.4	10.1	11.0	13.0	16.7	20.7	22.9	22.4	19.1	15.7	12.6	10.6	(19)	
		MCWB	9.3	8.7	9.1	9.5	12.2	15.1	16.8	16.8	15.6	13.6	10.9	9.4	(20)	
	2%	DB	9.4	9.0	9.6	11.4	15.0	17.9	20.3	20.6	17.7	14.3	11.2	9.7	(21)	
		MCWB	8.5	7.9	8.0	8.7	11.4	14.0	16.1	16.3	14.7	12.3	9.9	8.8	(22)	
	5%	DB	8.8	8.2	8.7	10.4	13.7	16.2	18.6	19.2	16.7	13.4	10.4	9.0	(23)	
		MCWB	8.0	7.1	7.3	8.2	10.8	13.0	15.1	15.7	14.1	11.6	9.4	8.1	(24)	
	10%	DB	8.1	7.5	8.1	9.6	12.7	15.0	17.4	18.0	15.9	12.6	9.8	8.3	(25)	
		MCWB	7.4	6.5	6.7	7.7	10.3	12.3	14.4	15.1	13.6	11.0	8.8	7.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	9.2	9.5	10.2	13.0	15.7	17.5	17.7	16.4	14.2	11.3	9.9	(27)	
		MCDB	10.1	9.6	10.3	11.6	15.7	19.5	21.6	20.8	18.1	15.3	12.0	10.3	(28)	
	2%	WB	8.8	8.2	8.4	9.4	12.0	14.4	16.4	16.8	15.4	13.0	10.3	9.0	(29)	
		MCDB	9.2	8.8	9.2	10.6	13.9	17.2	19.6	19.5	16.8	13.8	10.9	9.4	(30)	
	5%	WB	8.1	7.3	7.7	8.7	11.2	13.5	15.5	16.1	14.7	12.1	9.6	8.3	(31)	
		MCDB	8.6	8.0	8.5	10.0	13.0	15.7	18.0	18.4	16.0	13.0	10.3	8.8	(32)	
	10%	WB	7.5	6.6	7.0	8.1	10.6	12.7	14.8	15.5	14.1	11.3	9.0	7.6	(33)	
		MCDB	8.0	7.4	7.9	9.3	12.3	14.5	16.9	17.5	15.4	12.2	9.7	8.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.7	4.1	4.7	4.9	4.7	4.5	4.6	4.7	4.6	4.6	4.2	3.9	(35)	
		MCDBR	4.0	4.4	5.1	5.3	5.8	6.3	6.2	6.3	5.2	5.1	4.4	4.0	(36)	
	5% WB	MCWBR	3.8	3.8	4.1	4.2	4.1	3.9	3.7	3.9	3.9	4.1	3.9	3.9	(37)	
		MCWBR	3.9	4.1	4.8	4.8	5.2	5.5	5.5	5.6	4.8	4.7	4.2	3.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.318	0.325	0.350	0.385	0.385	0.373	0.375	0.373	0.365	0.351	0.327	0.309	(40)	
		taud	2.369	2.462	2.402	2.296	2.359	2.442	2.479	2.497	2.486	2.412	2.323	2.316	(41)	
	Edn at Noon	Edn at Noon	657	787	836	845	865	879	870	852	813	735	626	583	(42)	
		Edh at Noon	58	72	92	116	115	107	101	94	85	74	60	51	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.57	1.41	2.52	3.95	4.92	5.15	4.86	4.12	2.87	1.60	0.77	0.47	(44)	
	RadStd	RadStd	0.07	0.22	0.28	0.20	0.38	0.38	0.47	0.33	0.28	0.19	0.08	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page